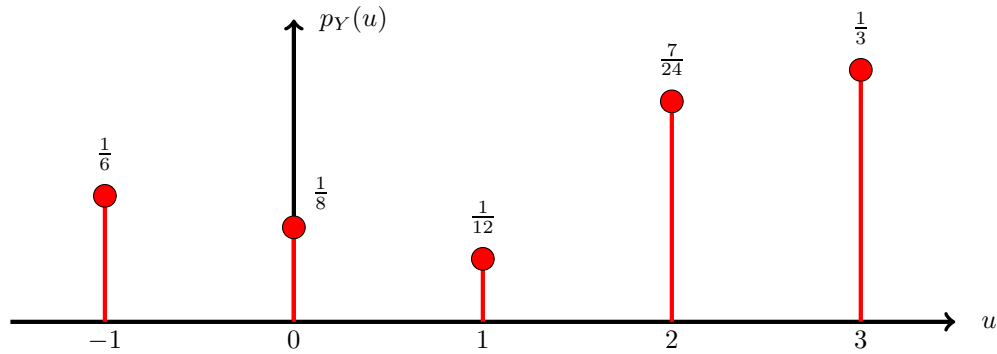


If a random variable Y has probability mass function shown below, then what is the probability that $Y^2 - 2Y - 3$ is nonnegative?

$$p(Y^2 - 2Y - 3) \geq 0$$



$$\begin{array}{r} -3 \\ 3 \\ -2 \end{array}$$

$$Y^2 - 2Y - 3 \geq 0$$

$$(Y-3)(Y+1) \geq 0$$

$$Y \leq -1 \cup Y \geq 3$$

$$\begin{aligned} \rightarrow P(Y^2 - 2Y - 3) &= P(Y = -1) + P(Y = 3) \\ &= \frac{3}{6} + \frac{1}{3} \end{aligned}$$

(a) $\frac{1}{2}$

(b) $\frac{1}{6}$

(c) $\frac{1}{4}$

(d) $\frac{1}{3}$

(e) $\frac{2}{3}$

(f) $\frac{1}{8}$

(g) $\frac{5}{8}$

(h) $\frac{3}{4}$

(i) $\frac{7}{8}$

(j) $\frac{7}{24}$

(k) $\frac{7}{12}$

(l) None of these

A random variable X has a probability density function which equals Ae^{-u} on the interval $[0, 1]$ and is zero otherwise, where A is a constant. What is the probability that X is less than $1/2$?

(a) $\frac{e-\sqrt{e}}{e-1}$

(b) $\frac{e-1}{e-1}$

(c) $\frac{e-\sqrt{e}}{e}$

(d) $\frac{e^2-e}{e-1}$

(e) $\frac{1}{e-1}$

(f) $\frac{e-\sqrt{e}}{e+1}$

(g) $\frac{e^2}{e-1}$

(h) $\frac{e}{e+1}$

(i) $\frac{e^2}{e+1}$

(j) $\frac{\sqrt{e}}{e-1}$

(k) $\frac{\sqrt{e}}{e+1}$

(l) $1/2$

(m) None of these

$$f_X(u) = Ae^{-u}$$

$$\int_0^1 Ae^{-u} du = 1$$

$$A[-e^{-u}]_0^1 = 1$$

$$A(-e^{-1} + 1) = 1$$

$$A(1 - \frac{1}{e}) = 1$$

$$A = \frac{1}{1 - \frac{1}{e}}$$

$$P(X < \frac{1}{2}) = \int_0^{\frac{1}{2}} Ae^{-u} du$$

$$= A(1 - e^{-\frac{1}{2}})$$

$$= \frac{1}{1 - \frac{1}{e}} (1 - e^{-\frac{1}{2}})$$

$$= \frac{(1 - e^{-\frac{1}{2}})e}{(1 - \frac{1}{e})e} = \frac{e - e^{\frac{1}{2}}}{e - 1} = \frac{e - \sqrt{e}}{e - 1}$$

Suppose a random variable X has a probability mass function which equals $p_X(k) = \frac{C}{2^k k!}$ for all integers $k \geq 1$, and equals zero elsewhere, where C is a constant. What is the probability that X^2 is less than three?

$$p_X(k) = \frac{C}{2^k k!} \quad \forall k \geq 1, k \in \mathbb{Z}, 0 \text{ elsewhere}$$

(a) $\frac{1/2}{\sqrt{e}-1}$

(b) $\frac{5/8}{\sqrt{e}-1}$

(c) $\frac{1}{\sqrt{e}-1}$

(d) $1/2$

(e) $1/4$

(f) $3/4$

(g) $\frac{1/2}{\sqrt{e}+1}$

(h) $\frac{1/2}{e-1}$

(i) $\frac{1/2}{e+1}$

(j) $\frac{2}{\sqrt{e}-1}$

(k) $\frac{1/4}{\sqrt{e}-1}$

(l) None of these

Poisson rv: $P(Y=k) = \frac{\lambda^k e^{-\lambda}}{k!}$

@ $\lambda = \frac{1}{2}$, $= \frac{(\frac{1}{2})^k e^{-\frac{1}{2}}}{k!}$

$$P(X=k) = P(Y=k | Y > 0)$$

$$= \frac{P(Y=k)}{P(Y > 0)}$$

$$P(Y > 0) = 1 - P(Y=0)$$

$$= \frac{P(Y=k)}{1 - P(Y=0)}$$

for $X^2 < 3$, $k=1$

$$\frac{(\frac{1}{2})^1 e^{-\frac{1}{2}}}{1!}$$

$$\rightarrow \frac{1 - \frac{(\frac{1}{2})^0 e^{-\frac{1}{2}}}{0!}}{1 - \frac{(\frac{1}{2})^0 e^{-\frac{1}{2}}}{0!}}$$

$$= \frac{\frac{1}{2} e^{-\frac{1}{2}} e^{\frac{1}{2}}}{1 - e^{-\frac{1}{2}} e^{\frac{1}{2}}} = \frac{\frac{1}{2}}{e^{\frac{1}{2}} - 1}$$